



Pharmacogenetic exploration of buprenorphine and related metabolites in umbilical cord blood

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ARTICLE INFO

Handling Editor: Prof. L.H. Lash

Keywords:

Pharmacogenetics
Polymorphism
Buprenorphine
Metabolites
Umbilical cord
Neonatal abstinence syndrome
Neonatal opioid withdrawal syndrome

ABSTRACT

The goal of this study was to explore associations between single-nucleotide polymorphisms (SNPs) and umbilical cord blood concentrations of buprenorphine and its metabolites following maternal administration. This is a sub-study of a prospective observational cohort investigation which included pregnant women receiving buprenorphine for opioid use disorder during pregnancy. Following delivery, umbilical cord blood samples were collected and genotyped using a pharmacogenetic panel. The drug and metabolite concentrations were analyzed through liquid chromatography-mass spectrometry, and genetic association analysis was completed using PLINK software. The included neonates ($n = 14$) had a mean birth weight of 3.00 ± 0.39 kg and were born to mothers receiving a mean buprenorphine dose of 10.29 ± 6.22 mg. Ten concentration groupings (drug, single metabolite, as well as drug/metabolite(s) combinations) produced 18 unique SNP associations. Two significant associations included variations in CYP3A4 and UGT1A1, which play a role in the metabolism of buprenorphine. These preliminary findings suggest potential pharmacogenetic factors influencing fetal drug exposure, warranting larger studies to validate associations and explore clinical implications for neonatal outcomes.

1. Introduction

Both misused prescription opioids and non-prescription opioid use remain a devastating public health issue in the United States and around the globe [1,2]. Further, opioid-related drug overdose contributed to over 80,000 deaths in 2022 [3]. However, this number declined to an estimated 54,000 in 2024, according to provisional data reported by the U.S. Centers for Disease Control and Prevention [4]. Opioid drug use, particularly amongst pregnant women aged 15–44 years, has increased from 5.4 % in 2019 to nearly 8.0 % in 2020 [5]. Between 1992 and 2012, the population of pregnant women who were using prescription opioids, as the main substance of abuse, and received treatment increased from 1 % to 19 %, and in that same period, the number of pregnant women seeking treatment for misuse of prescription opioids increased by 26 % [6].

With increasing rates of opioid use disorder among pregnant women, neonatal abstinence syndrome (NAS), also known as neonatal opioid withdrawal syndrome (NOWS) [7], has become a significant public

health concern [8]. NAS is an umbrella term encompassing multiple conditions resulting from fetal exposure to drugs, such as opioids, *in utero*. Opioid affected infants experience physiological dependence and exhibit symptoms such as high-pitched continuous crying, difficulty feeding, respiratory issues, seizures, and severe irritability. Prevention of NAS and associated neonatal deaths could ideally be accomplished through abstinence from drug use [9]; however, medication-assisted therapy [10], historically methadone [11] or buprenorphine (BUP) [12], is often required owing to maternal drug dependency. Further, with proper adherence, the transfer of withdrawal symptoms to the fetus may be mitigated. Methadone rose as the standard treatment for opioid dependency in the late 1900s due to its effectiveness and ability to increase birth weight [13]; however, BUP has gained favor due to advantages related to increased birth weight and decreased chance of preterm birth [14].

Genetic variations can have a significant role in the expression and function of many transporters, receptors, and metabolizing enzymes, including cytochrome P450 (CYP) and uridine 5'-diphospho-

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<https://doi.org/10.1016/j.toxrep.2025.102093>

Received 22 April 2025; Received in revised form 18 June 2025; Accepted 19 July 2025

Available online 20 July 2025

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glucuronosyltransferase (UGT) enzymes, and consequently, the response to drugs [15–17]. Characterizing these variations and applying them clinically may have a profound effect on improving neonatal outcomes [18]. NAS is reported to be a polygenic disorder based on the data associated with the genes relevant to opioids [19]. A number of single-nucleotide polymorphisms (SNPs) are reported to influence the treatment and hospital stay length for neonates.

Several CYP enzymes produce BUP metabolites, including norbuprenorphine (NBUP), which is a potent opioid agonist and primarily produced via N-dealkylation through CYP3A4 and CYP3A5 [20,21]. A case report studying the functionality of CYP3A4 has supported the idea of CYP3A4*1B being associated with ultra-rapid metabolizers [2]. The authors stated that having at least one copy of the CYP3A4*1B will cause a rapid metabolism of BUP relative to an extensive metabolizer. As such, individuals carrying the CYP3A4*1B allele may require BUP dose adjustment to ensure opioid use disorder (OUD) treatment efficacy.

BUP is subject to metabolism by UGT as well [16]. Both BUP and NBUP are glucuronidated by UGT1A1, UGT1A3, and UGT2B7. In an *in vitro* study examining liver enzymes, researchers found that the presence of allele UGT1A1*28 contributed to a decrease in BUP glucuronidation, while the UGT2B7 promoter (G-842A) mutation showed a significant increase in BUP glucuronidation. A study concerning the binding affinity and pharmacological activity of BUP metabolites, buprenorphine-3-glucuronide (BUP-Gluc) and norbuprenorphine-3-glucuronide (NBUP-Gluc), revealed distinct pharmacological effects of both metabolites, each yielding an opiate effect [22].

The utilization of pharmacogenomic information in predicting safe and efficacious use of BUP has been reported in the literature [23,24]. In this sub-study, we explored the associations between various SNPs and umbilical cord blood BUP, metabolite, and combination concentrations using samples obtained from a previous study [25].

2. Materials and methods

2.1. Clinical sampling

The original study [25] was conducted as a prospective observational cohort study on the use of BUP in pregnant women. Expectant mothers receiving BUP-based treatment, Subutex® or Suboxone®, were identified and evaluated for inclusion criteria (gestational age of ≤ 36 weeks at enrollment, and no polysubstance abuse, or congenital abnormalities). Those who met the inclusion criteria provided written informed consent before participating in the study. The study was approved by the East Tennessee State University/Veterans Affairs Medical Campus Institutional Review Board (IRB#: 0612.8 s). After delivery, demographic data were collected, and cord blood samples were obtained and stored. Neonates were monitored and treated for NAS according to institutional protocols.

2.2. Analysis of cord blood samples

Samples underwent protein precipitation and solid phase extraction, then liquid chromatography-mass spectrometry (LC-MS) to determine BUP and metabolite concentrations [25]. The LC-MS method was validated with inter- and intra-day accuracy and precision $< 15\%$. Aliquots of the collected blood were genotyped as described in a previous study [26,27]. Briefly, the whole blood was sent to Vanderbilt Technologies for Advanced Genomics (VANTAGE) for DNA extraction and sequencing using Sequenom iPLEX PGx Pro Panel v1.0 (Sequenom, San Diego, CA, USA) for genetic analysis. The MassARRAY Typer software was utilized for determining genotype call rates.

2.3. Statistical analysis

An additive genetic association model was utilized, with the minor allele acting as the dosage. Multiple linear regression for continuous

traits was used to determine associations between SNPs and concentrations of interest. The analyses were adjusted for the covariates of birth weight and maternal BUP dose. PLINK v1.09 was used to estimate minor allelic frequency, Hardy-Weinberg equilibrium, regression coefficients, 95 % confidence intervals, and *p*-values [28]. Continuous data are shown as mean $\pm$ standard deviation. Statistical significance was defined as $p < 0.05$ for this study. Given the exploratory nature and small sample size of this study, formal power calculations were not feasible. However, we acknowledge that the lack of correction for multiple testing increases the risk of Type I error, and these findings should be considered preliminary.

3. Results

Although 19 umbilical cord blood samples were collected in the parent study [25], only 14 samples were viable for genotyping in this sub-study. As shown in Table 1, the mean birth weight of included neonates was 3.00 ± 0.39 kg. Mean BUP and metabolite concentrations are shown in Table 2. BUP ($n = 14$), BUP-Gluc ($n = 13$), NBUP ($n = 14$), and NBUP-Gluc ($n = 14$) were found to have mean concentrations of 6.43 ± 4.76 ng/mL, 15.38 ± 8.19 ng/mL, 15.65 ± 7.48 ng/mL, and 25.77 ± 17.88 ng/mL, respectively. Additionally, six combinations of BUP and/or metabolite concentrations were assessed.

Significant SNP-concentration associations are shown in Table 3. BUP concentration showed a significant association with two SNPs, rs1080985 and rs3213619. While BUP-Gluc was associated with eight SNPs (rs4149117 and rs7311358 in *SLCO1B3*; rs7294, rs9934438, and rs9923231 within *VKORC1*; rs1801265 in *DPYD*; rs4124874 in *UGT1A1*; and rs628031 in *SLC22A1*). NBUP by itself was found to be associated with two SNPs (rs55785340 found on *CYP3A4* with a positive association ($\beta = 14.47$), while rs2273697 found on *ABCC2* had a negative association ($\beta = -6.77$). Finally, two SNPs (rs3740066 and rs1143672) in the *ABCC2* and *SLC15A2* genes, respectively, were found to be associated with NBUP-Gluc.

This sub-study also examined various combined drug and metabolite concentrations, as seen in Table 4. The BUP + BUP-Gluc combination values were found to be associated with six SNPs, most of which were also associated with BUP-Gluc, except rs2273697. Four SNPs, rs55785340, rs4244285, rs35742686, and rs762551, were associated with BUP + NBUP concentration, all of which had positive beta values. Of note, only one of these four SNPs, rs55785340, was associated with a single metabolite concentration, NBUP. BUP-Gluc + NBUP-Gluc was also associated with eight SNPs, two of which were associated with negative beta values (rs1801265 and rs3740066). Of these eight SNPs, one novel SNP was identified, rs3745274 on *CYP2B6*. The combination of NBUP + NBUP-Gluc was significantly associated with only one SNP, rs4244285 found on *CYP2C19*, which had a positive beta value, 21.00.

Additionally, three metabolites were analyzed together, BUP-Gluc + NBUP + NBUP-Gluc, which produced one SNP association. In that case, a positive beta (15.66) was returned for rs1143672 found on *SLC15A2*, although not unique to this combination. The four drug/metabolites combination, BUP + BUP-Gluc + NBUP + NBUP-Gluc, produced a cumulative eight SNPs with which it was significantly associated. None of these SNPs were unique to this analysis and possessed mostly positive beta values, with the exceptions of rs1801265 and rs3740066, -18.47 and -21.63 , respectively.

Table 1
Subject demographics.

Parameter	N	Values (mean $\pm$ SD)
Birth weight (kg)	14	3.00 ± 0.39
Maternal BUP dose (mg)	14	10.29 ± 6.22

BUP, buprenorphine; SD, standard deviation.

Table 2

Concentrations of BUP, metabolites, and combinational groupings in umbilical cord blood.

Drug and/or metabolite(s)	N	Concentration (ng/mL) (mean $\pm$ SD)
BUP	14	6.43 $\pm$ 4.76
BUP-Gluc	13	15.38 $\pm$ 8.19
NBUP	14	15.65 $\pm$ 7.48
NBUP-Gluc	14	25.77 $\pm$ 17.88
BUP + NBUP	14	22.08 $\pm$ 8.28
BUP-Gluc + NBUP-Gluc	13	41.80 $\pm$ 24.32
BUP + BUP-Gluc	13	22.24 $\pm$ 11.29
NBUP + NBUP-Gluc	14	41.42 $\pm$ 19.87
BUP-Gluc + NBUP + NBUP-Gluc	13	57.59 $\pm$ 25.21
BUP + BUP-Gluc + NBUP + NBUP-Gluc	13	64.45 $\pm$ 28.51

BUP, buprenorphine; BUP-Gluc, buprenorphine-glucuronide; NBUP, norbuprenorphine; NBUP-Gluc, norbuprenorphine-glucuronide; SD, standard deviation.

4. Discussion

This sub-study aimed to identify pharmacogenetic associations between cord blood concentrations of BUP, its metabolites, and combinations thereof in infants exposed to the drug before birth.

The association between BUP concentration and rs1080985 in *CYP2D6* represents a novel finding (Table 3). In a previous study, BUP and its major metabolite, NBUP, were found to be *in vitro* inhibitors of CYP2D6; however, at therapeutic concentrations, they are not associated with clinically important interactions [29]. This SNP, being in the promoter region of CYP2D6, has been shown to yield higher gene expression and, thereby, more enzymatic activity [30]. The observed negative association suggests that more doses of the minor allele G appear to correlate with reduced BUP concentration. While CYP2D6 is not a metabolic pathway for BUP, indirect interactions remain possible. As such, no studies to date have linked rs1080985 and BUP.

The second SNP linked with BUP in our sub-study was rs3213619 in *ABCB1*. This variant is associated with increased P-gp expression [31] which acts as an efflux transporter that actively pumps opioids out of cells in the blood-brain barrier, potentiating suboptimal pain relief [32]; however, as BUP is not known to be influenced by P-gp activity, using *in vitro* and *in vivo* assays [33]. A recent study has shown an inverse relationship between *ABCB1* placental methylation and the severity of NOWS [34]. Currently, no other studies have suggested an association of rs3213619 and BUP fetal cord blood concentration.

This study also looked at the active metabolites of BUP, including

BUP-Gluc. BUP-Gluc was significantly associated with 8 different SNPs, mainly from the genes *SLOC1B3* and *VKORC1*. To date, there have been no studies that show an association between *SLOC1B3* and BUP-Gluc or NAS; however, SNPs in *SLOC1B3* have been implicated in an increased risk of acute rejection in lung allograft recipients in patients receiving mycophenolic acid [35]. Also, rs4149056 in *SLCO1B1* is shown to be a strong risk factor for the development of muscle symptoms in patients who are on statins [36,37]. As for the three SNPs on *VKORC1*, a gene widely implicated in warfarin dosing, no studies to date have been conducted on the relationship between *VKORC1* SNPs and BUP or its metabolites.

Another SNP associated with the metabolite BUP-Gluc is rs1801265 from the *DPYD* gene. This mutation causes a change in amino acids, which can affect enzymatic activity and is typically utilized in predicting severe toxicity in chemotherapeutic drugs [38]. No existing literature exists on the correlation between *DPYD* or its corresponding SNP with BUP or related metabolites. In addition, BUP-Gluc was positively associated with the SNP rs4124874 from the *UGT1A1* gene. Although this specific SNP has yet to be reported on in this context, *UGT1A1* has a significant role in the glucuronidation of BUP [16]. Another SNP associated with BUP-Gluc in this study was rs628031 from *SLC22A1*. Although no evidence was found linking this gene to BUP, a pharmacogenomic study conducted in 2016 reported that *SLC22A1* was susceptible to genetic variability in infants treated with tramadol [39]. The *SLC22A1* gene, which encodes organic cation transporter 1, was found to influence the pharmacokinetics of tramadol and, eventually, the concentrations of O-desmethyl tramadol, the active metabolite. In addition, polymorphism of *SLC22A1* is linked to altered levels of tramadol metabolites and elevated risk of toxicity [40].

Two SNPs, rs55785340 and rs2273697, were found to be associated with the metabolite NBUP; however, there is no existing literature relating these SNPs and this metabolite concentration. While the individual SNP, rs55785340, has no studies on its role in BUP, the SNP has been associated with higher systemic exposure to simvastatin [41]. In prior studies, rs2273697 has been implicated in drug resistance to some antiepileptic drugs in various study populations [42–45]. Another study of several SNPs in *ABCC2*, including rs2273697, conducted in Japan revealed a significant association ($p = 0.011$) between genotype AA at position 1249 and an elevated risk of kidney tubular dysfunction in patients with HIV-1 treated with tenofovir, suggesting that this SNP may play a role in developing kidney tubular dysfunction during that particular treatment [46].

Table 3

SNPs significantly associated with drug or metabolite concentration groupings.

Drug or metabolite	Gene	SNP	BP ^a	A1 ^b	MAF ^c	HWE ^d	Beta (95 % CI) ^e	p-value
BUP	<i>CYP2D6</i>	rs1080985	42,528,382	G	0.143	1.000	−6.23 (−10.33 to −2.12)	0.016
	<i>ABCB1</i>	rs3213619	87,230,193	C	0.167	1.000	6.47 (2.01–10.93)	0.017
BUP-Gluc	<i>SLCO1B3</i>	rs4149117	21,011,480	T	0.167	1.000	10.82 (4.14–17.50)	0.011
	<i>SLCO1B3</i>	rs7311358	21,015,760	G	0.167	1.000	10.82 (4.14–17.50)	0.011
	<i>VKORC1</i>	rs7294	31,102,321	A	0.433	0.615	−7.97 (−14.23 to −1.71)	0.034
	<i>DPYD</i>	rs1801265	98,348,885	C	0.267	0.226	−6.16 (−11.01 to −1.31)	0.035
	<i>VKORC1</i>	rs9934438	31,104,878	A	0.267	0.507	−12.68 (−22.90 to −2.47)	0.038
	<i>VKORC1</i>	rs9923231	31,107,689	T	0.267	0.507	−12.68 (−22.90 to −2.47)	0.038
	<i>UGT1A1</i>	rs4124874	234,665,659	C	0.467	0.326	7.62 (1.27–13.97)	0.043
	<i>SLC22A1</i>	rs628031	160,560,845	A	0.367	1.000	−6.06 (−11.14 to −0.97)	0.045
NBUP	<i>CYP3A4</i>	rs55785340	99,365,983	G	0.038	1.000	14.47 (3.36–25.58)	0.031
	<i>ABCC2</i>	rs2273697	101,563,815	A	0.250	1.000	−6.77 (−12.53 to −1.01)	0.044
NBUP-Gluc	<i>ABCC2</i>	rs3740066	101,604,207	A	0.286	0.258	−14.43 (−25.13 to −3.73)	0.025
	<i>SLC15A2</i>	rs1143672	121,648,168	A	0.464	0.316	10.01 (1.63–18.39)	0.041

SNP, single nucleotide polymorphism; BP, base pair; BUP, buprenorphine; BUP-Gluc, buprenorphine-glucuronide; NBUP, norbuprenorphine; NBUP-Gluc, norbuprenorphine-glucuronide.

^a Physical position.

^b Minor allele.

^c Minor allele frequency overall.

^d p-value for Hardy-Weinberg equilibrium test.

^e Beta adjusted for covariates (birth weight and maternal drug dose).

Table 4
SNPs significantly associated with drug and/or metabolite combinational concentrations.

Drug and/or metabolites	Gene	SNP	BP ^a	A1 ^b	MAF ^c	HWE ^d	Beta (95 % CI) ^e	p-value
BUP + BUP-Gluc	<i>VKORC1</i>	rs7294	31,102,321	A	0.433	0.615	−11.97 (−19.49 to −4.44)	0.012
	<i>SLCO1B3</i>	rs4149117	21,011,480	T	0.167	1.000	13.74 (4.41–23.08)	0.018
	<i>SLCO1B3</i>	rs7311358	21,015,760	G	0.167	1.000	13.74 (4.41–23.08)	0.018
	<i>UGT1A1</i>	rs4124874	234,665,659	C	0.467	0.326	11.06 (3.11–19.01)	0.023
	<i>ABCC2</i>	rs2273697	101,563,815	A	0.233	1.000	9.36 (2.07–16.64)	0.033
BUP + NBUP	<i>DPYD</i>	rs1801265	98,348,885	C	0.267	0.226	−7.96 (−14.55 to −1.36)	0.042
	<i>CYP3A4</i>	rs55785340	99,365,983	G	0.036	1.000	21.09 (9.41–32.77)	0.006
	<i>CYP2C19</i>	rs4244285	96,541,616	A	0.143	1.000	10.67 (2.58–18.76)	0.029
	<i>CYP2D6</i>	rs35742686	42,524,244	D	0.033	1.000	20.30 (3.85–36.75)	0.036
	<i>CYP1A2</i>	rs762551	75,041,917	C	0.400	1.000	8.01 (1.28–14.75)	0.042
BUP-Gluc + NBUP-Gluc	<i>DPYD</i>	rs1801265	98,348,885	C	0.267	0.326	−16.68 (−28.89 to −4.47)	0.025
	<i>SLC15A2</i>	rs1143672	121,648,168	A	0.467	0.604	15.47 (4.07–26.87)	0.026
	<i>ABCC2</i>	rs3740066	101,604,207	A	0.300	0.539	−19.35 (−33.94 to −4.75)	0.029
	<i>UGT1A1</i>	rs4124874	234,665,659	C	0.467	0.326	20.70 (4.71–36.7)	0.032
	<i>SLCO1B3</i>	rs4149117	21,011,480	T	0.167	1.000	24.73 (5.33–44.13)	0.034
	<i>SLCO1B3</i>	rs7311358	21,015,760	G	0.167	1.000	24.73 (5.33–44.13)	0.034
	<i>CYP2B6</i>	rs3745274	41,512,841	G	0.467	0.326	20.24 (3.72–36.77)	0.040
	<i>ABCC2</i>	rs2273697	101,563,815	A	0.233	1.000	17.51 (2.93–32.09)	0.043
	<i>CYP2C19</i>	rs4244285	96,541,616	A	0.154	1.000	21.00 (7.19–35.10)	0.020
	<i>SLC15A2</i>	rs1143672	121,648,168	A	0.467	0.604	15.66 (2.94–28.37)	0.039
NBUP + NBUP-Glu								
BUP-Gluc + NBUP + NBUP-Gluc	<i>UGT1A1</i>	rs4124874	234,665,659	C	0.467	0.326	24.14 (5.76–42.53)	0.030
	<i>SLC15A2</i>	rs1143672	121,648,168	A	0.467	0.604	17.40 (3.95–30.86)	0.032
BUP + BUP-Gluc + NBUP + NBUP-Gluc	<i>DPYD</i>	rs1801265	98,348,885	C	0.267	0.226	−18.47 (−33.05 to −3.90)	0.035
	<i>ABCC2</i>	rs2273697	101,563,815	A	0.233	1.000	20.84 (4.28–37.39)	0.036
	<i>CYP2B6</i>	rs3745274	41,12,841	G	0.467	0.326	23.72 (4.77–42.67)	0.037
	<i>ABCC2</i>	rs3740066	101,604,207	A	0.300	0.539	−21.63 (−38.92 to −4.34)	0.037
	<i>SLCO1B3</i>	rs4149117	21,011,480	T	0.167	1.000	27.65 (4.71–50.58)	0.042
	<i>SLCO1B3</i>	rs7311358	21,015,760	G	0.167	1.000	27.65 (4.71–50.58)	0.042

SNP, single nucleotide polymorphism; BP, base pair; BUP, buprenorphine; BUP-Gluc, buprenorphine-glucuronide; NBUP, norbuprenorphine; NBUP-Gluc, norbuprenorphine-glucuronide.

- ^a Physical position.
- ^b Minor allele.
- ^c Minor allele frequency overall.
- ^d p-value for Hardy-Weinberg equilibrium test.
- ^e Beta adjusted for covariates (birth weight and maternal drug dose).

Additionally, in this sub-study, the metabolite NBUP-Gluc was found to be associated with rs1143672, which also does not have any existing literature on its connection with BUP or its metabolites. Prior studies on the Han Chinese population reveal that rs1143672 in *SLC15A2* has been correlated with warfarin dosing, suggesting that those subjects carrying rs1143672 should take a lower dose of warfarin [47]. Another variation of *SLC15A2*, rs2257212, is reported to increase the risk of renal toxicity associated with colistin [48]. Further, rs3740066 does not appear to have an existing link within the literature.

In addition, the current sub-study also looked at the effects of combined metabolites and their associated SNPs (Table 4). In this sub-study, four additional SNPs were identified that were not seen in single metabolite concentrations, and to date, there is no existing literature revealing any association with BUP metabolites. BUP + NBUP produced three unique, significantly associated SNPs, rs4244285 (also associated with NBUP + NBUP-Gluc), rs35742686, and rs762551. The first SNP, rs4244285, found on *CYP2C19* has been correlated with significantly higher plasma levels of a methadone metabolite, ethylidene-1,5-dimethyl-3,3-diphenylpyrrolidene [49]. Another study has found that the polymorphism rs35742686 was not associated with increased OUD risks [50].

Further, BUP-Gluc + NBUP-Gluc produced one unique SNP, rs3745274 (which was also seen in BUP + BUP-Gluc + NBUP + NBUP-Gluc). Two genotypes of rs3745274 (GG and GT) have been associated with a higher risk of addiction in methadone responders, as well as addiction relapse rates in methadone non-responders [51]. Another study found that rs3745274 (*CYP2B6*:c.516 G>T polymorphism) was associated with a reduction in the metabolism of propofol in a cohort of 108 patients receiving anesthesia [52].

This study had several limitations of note, starting with the small sample size (n = 14). Due to the exploratory nature of this study, there was not a suitable population of subjects to attain statistical power, thus limiting the generalizability. In addition, this study was conducted on an ethnically homogenous population, making replication a challenge while potentially weakening its validity in broader populations. Additionally, the functional immaturity of neonatal drug-metabolizing enzymes and transporters compared to adults represents a fundamental difference that could influence the clinical relevance of these associations. Future studies should correlate these pharmacogenetic associations with clinical outcomes, including NAS severity scores, length of hospital stay, and need for pharmacological treatment.

5. Conclusion

This exploratory study identified multiple SNP associations between BUP and metabolites in umbilical cord blood, highlighting the potential role of pharmacogenetic variation in fetal drug exposure. Several unique SNPs were identified in association with the various combinational groupings. Some of the significant associations involved *CYP3A4* and *UGT1A1* variants, consistent with known metabolic pathways. Understanding these SNPs may help navigate the plethora of clinical presentations of NAS. Future studies with larger sample sizes and more comprehensive genetic interrogation, such as transcriptome- or genome-wide association studies, are needed to confirm our exploratory findings. These associations may be useful in creating future treatment directions and improving health outcomes for neonates exposed to BUP *in utero*.

CRediT authorship contribution statement

Darshan S. Shah: Writing – review & editing, Data curation, Conceptualization. **Derek E. Murrell:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Stacy D. Brown:** Writing – review & editing, Data curation. **Melissa Hoang:** Writing – review & editing, Writing – original draft, Data curation. **Sam Harirforoosh:** Writing – original draft, Supervision, Data curation, Conceptualization. **Amelia Monfared:** Writing – review & editing, Writing – original draft, Data curation.

Author statement

We confirm that the manuscript has been read and approved by all named authors.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors did not use any AI-assisted technology.

Funding

This study was partly funded by a Research and Development Committee Interdisciplinary Grant from East Tennessee State University.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was done at East Tennessee State University. The Vanderbilt VANTAGE Core provided technical assistance for this work. VANTAGE is supported in part by CTSA Grant (5UL1 RR024975-03), the Vanderbilt Ingram Cancer Center (P30 CA68485), the Vanderbilt Vision Center (P30 EY08126), and NIH/NCRR (G20 R030956). We would like to thank Drs. Cara Carter, Lauren Dickerson, and Russell Fankhouser for their contribution to this sub-study.

Data availability

Data will be made available on request.

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